Numerical Programming

CP1: edge detection, object detection, movement speed

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Chapter 1

Introduction

The goal of the project is to implement object detection using edge detection and algorithms.

Chapter 2

The algorithms used

First I extract frames from the video using OpenCV. Then I split it into three channels: red, green and blue. This helps with colors that would otherwise be lost due to convertion to grayscale. After comes preprocessing of the frames which uses edge-detection kernel and frame difference with couple others to better isolate the area of the moving object.
Now comes the algorithm that accepts the thresholded image and clusters it. For clustering I used Region growing algorithm which is simple region-based image segmentation method and does similar work as other data clustering algorithms. This algorithm goes over pixels of the image and visit's each pixel's neighbours and adds them to the region accordingly. After visiting all pixels we have image segmented into regions which denote moving objects in the scene. Tracking of the objects is done via calculating centroids from the regions, which then are used to calculate velocity by observing distances between new and old centroids over time.

Image interpolation algorithms are used for scaling down image for faster computation.

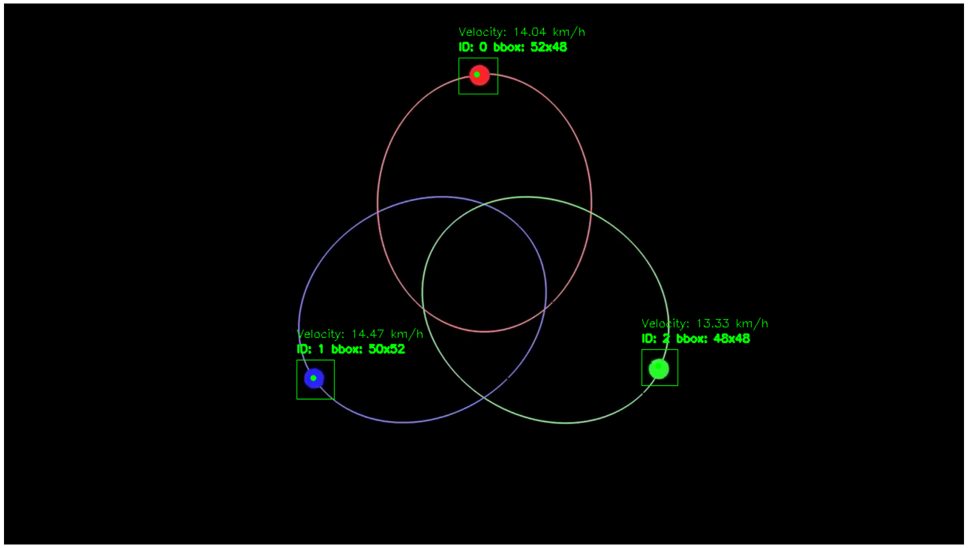
Chapter 3

Selected Demos

I have selected three videos for the testing purposes.

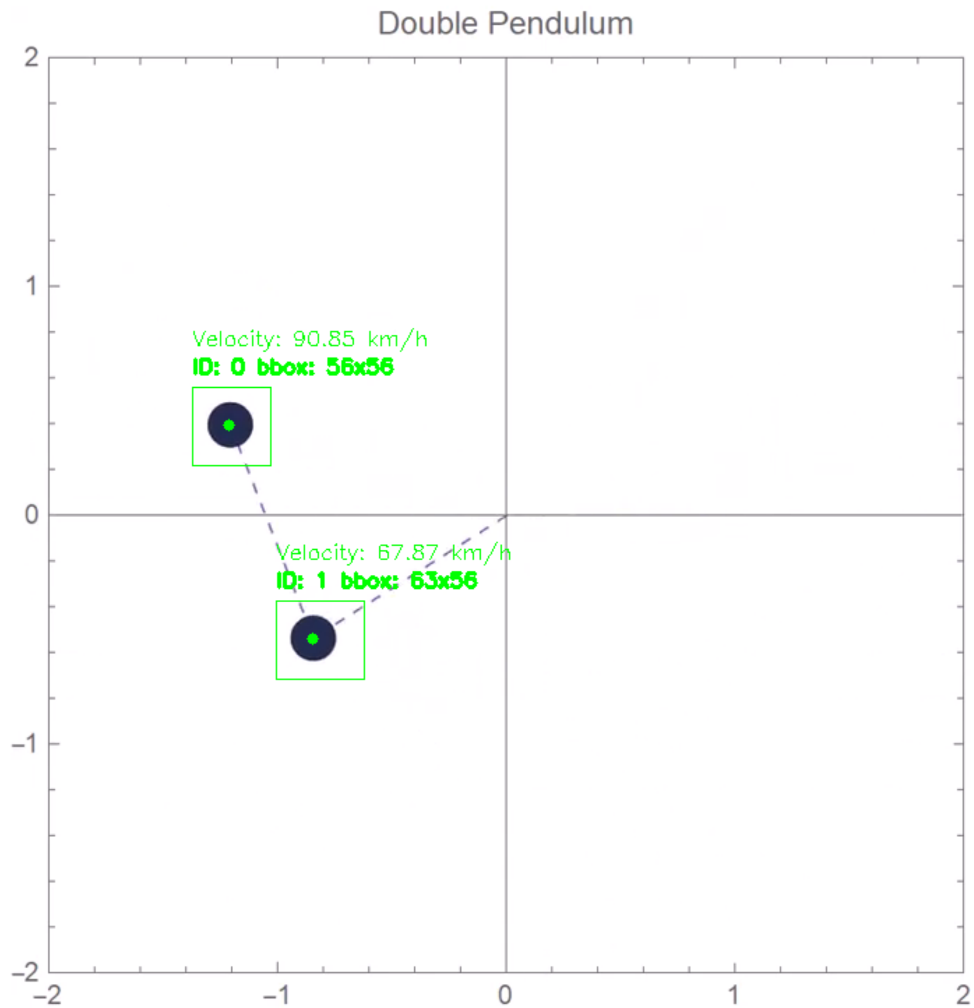
3.1 Three body problem simulation

Here the object detection works great and it keeps track of all three circles without any noticable artifacts. Though we can see that algorithm sometimes struggles with identifiying which ball is which.



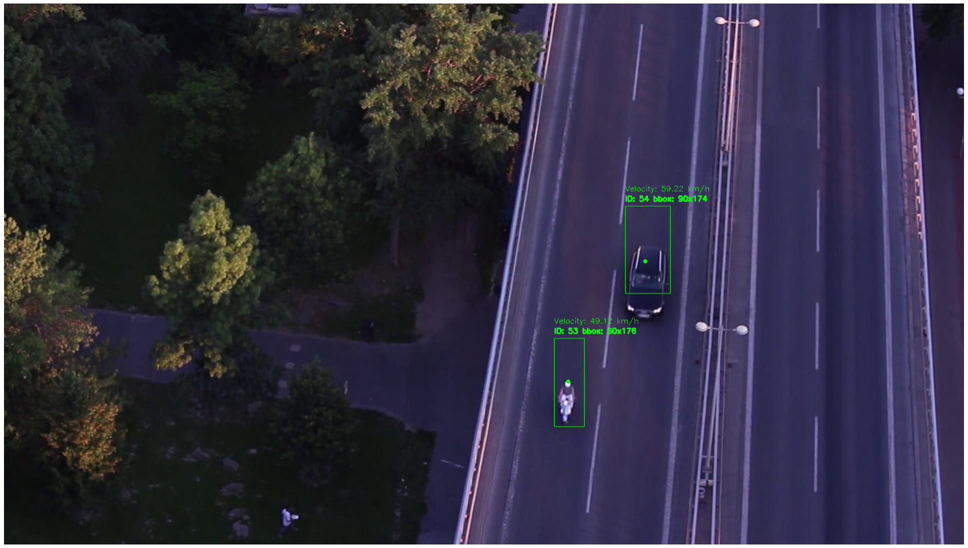
3.2 Dual pendulum

Here the algorithm has no issues at all with both identifying and distinguishing objects from each other.



3.3 Traffic

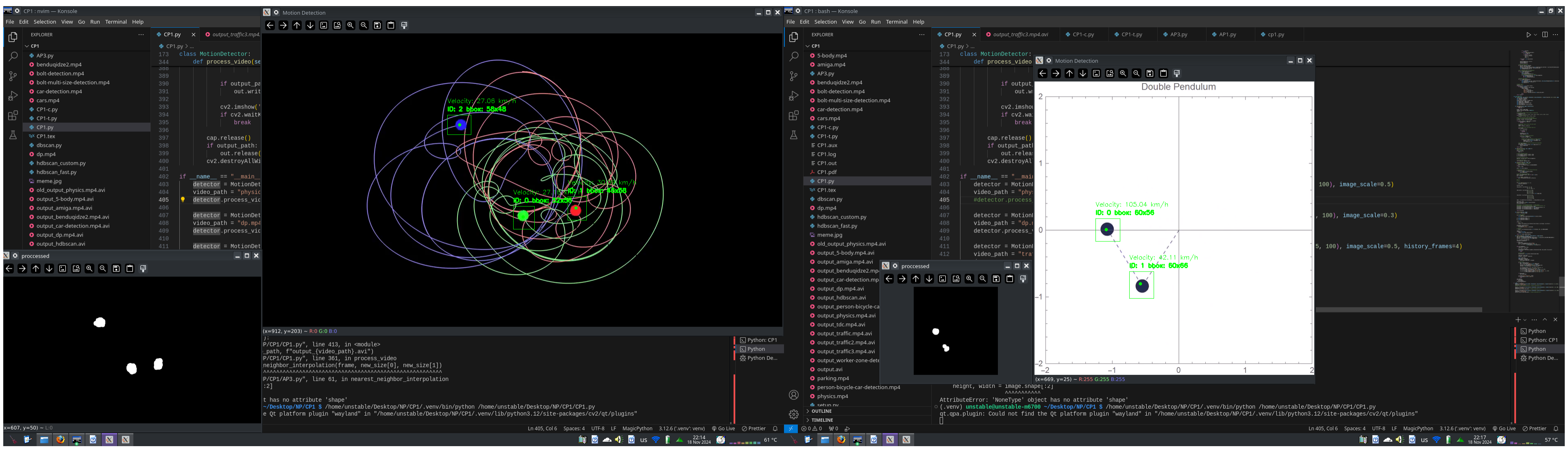
Here we see how algorithm tackles the real-world footage. It seems to be pretty stable all the vehicles are identified and kept track of with according velocities. However there are sometimes issues with slight movements like shadows on the trees but it handles well breeze. There can also be noted that black or long vehicles are harder to keep track of due to implementation of the object detection. Since we work with red, green and blue colors, black carries little information but it is still tracked pretty well.



Chapter 4

Encountered issues

As we discussed eralier, the issues are problems with objects overlapping and losing tracking (though, algorithm does keep track of them after seperation), camera distorshion and shaking causes massive fragments in image which can be mitigated by camera smoothing and noise reducing algorithms, black objects being hard to track and sometimes keep jittering due to lack of information. Below is demonstrated how the alogrithm sees the frame after it has been preprocessed.



In the small window in the left bottom corner is what is segmented via algorithm, which makes it impossible to distinguish between colors or objects overlapping since it’s only white. This can be improved by introducing color and other biases to the algorithm so that it can absorb more information from the frame and distinguish objects better.

Chapter 5

Conclusion

Object recognition with edge detecting is fascinating and very difficult to do, aspecially with the language such as python. With more time this program can be reimplemented with solutions discussed above in other languages which offer better computational speed and robustness. Nowadays object recognition also uses Maching Learning and IoT technologies to better keep track and even recognize objects.